Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Driver 2 and matrix

File: matrix2.kicad_sch

Driver 3 and matrix

File: matrix3.kicad_sch

Driver 4 and matrix

File: matrix4.kicad_sch

Driver 5 and matrix

File: matrix5.kicad_sch

Rails: VBUS -> VSYS 3.0–4.5 V (LED SW lines) -> +3V3 -> +1V1 at the RP2350.
5x IS31FL3745-CLS4-TR WLCSP-36 drivers (47 RGB LEDs each = 235 total, zero exposed ground pads).
Nets between sheets are global labels; rails are power symbols.
Requirements: design/intent.md. Parts: design/parts/drivers-bga.md. Decisions: design/decisions.md.

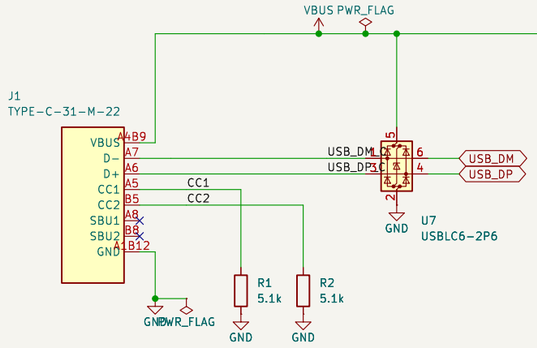
235 RGB LEDs on 38 mm round PCB (5x IS31FL3745, USB-C, Batt, RTC)
Variant f-usbc-rtc from f-usbc (inward 1.25 mm BT1 + RX8130CE RTC)
CERN-OHL-P-2.0

Sheet: /
File: round-pendant.kicad_sch

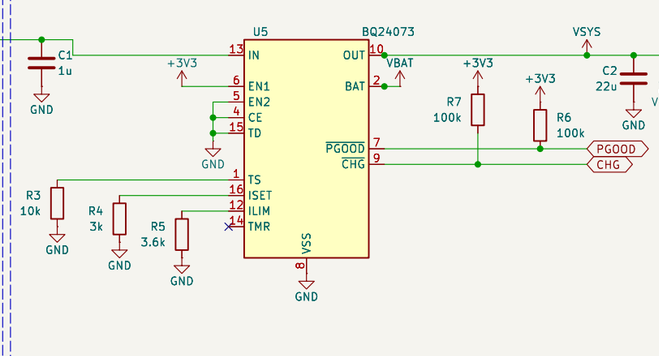
Title: round-pendant: top

Size: A4	Date: 2026-11-09	**Rev: FUR**
KiCad E.D.A. 9.0.8		Id: 1/8

USB-C 16P SMT & ESD

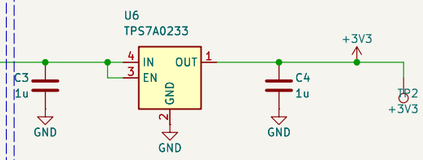


CHARGER AND POWER PATH



USB500: 450–500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

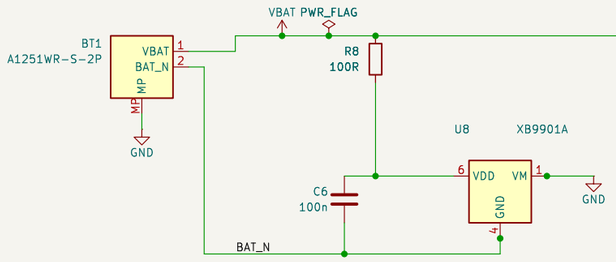
3V3



3V3 only ever follows VSYS down: the driver inputs
stay within VCC + 0.3 V (drivers-bga.md). 150 mA max.

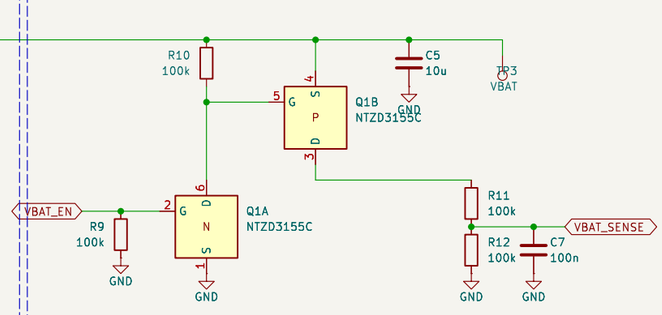
CELL AND PROTECTION

Inward 1.25 mm 2P SMT socket (A1251WR-S-2P, C225111); verify harness polarity (+ pin 1, − pin 2).



BAT_N: cell minus. Pins 4, 5 and the pad. Never pour GND here,
or protection is bypassed (protection.md).

VBAT SENSE



Zero draw off. ADC reads true only while 3V3 regulates.
VBAT >= 3.5 V (intent R22). Wait >= 5 ms after enabling.

USB-C 16P SMT, 1.25 mm battery socket, charger, protection, LDO
Variant f-usbc-rtc (BT1 swapped to inward 1.25 mm A1251WR-S-2P)
CERN-OHL-P-2.0

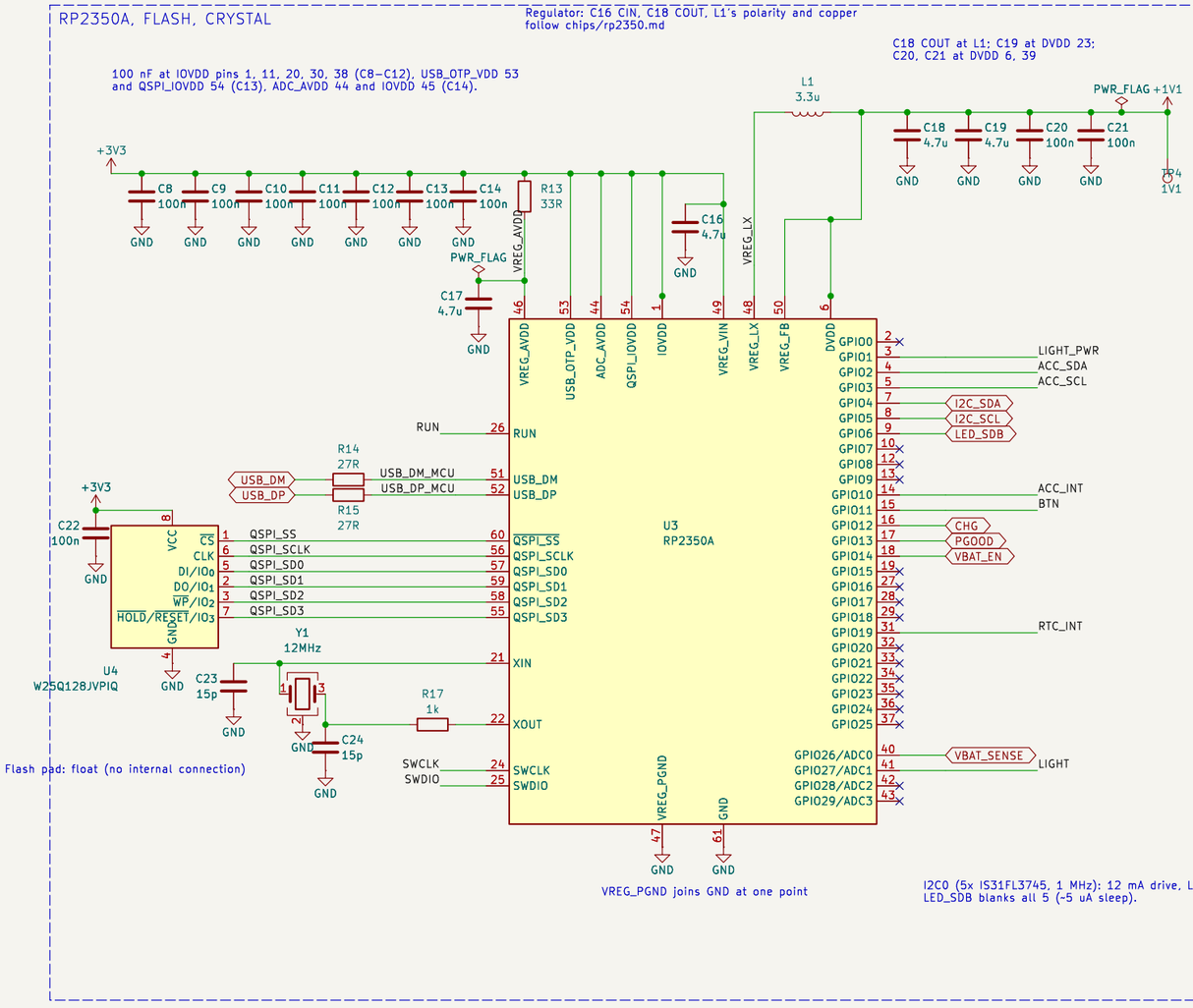
Sheet: /Power/		
File: power.kicad_sch		
Title: round-pendant: power		
Size: A3	Date: 2026-11-09	**Rev: FUR**
KiCad E.D.A. 9.0.8		Id: 2/8

RP2350A, FLASH, CRYSTAL

Regulator: C16 CIN, C18 COUT, L1's polarity and copper follow chips/rp2350.md

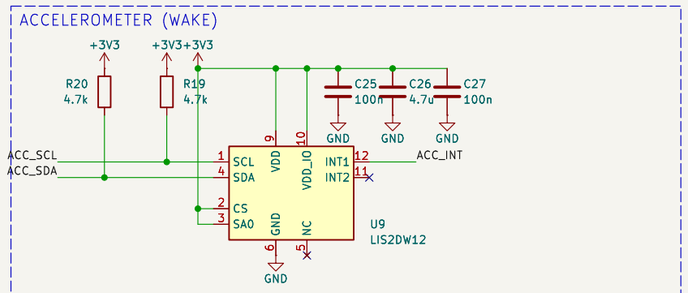
C18 COUT at L1; C19 at DVDD 23; C20, C21 at DVDD 6, 39

100 nF at IOVDD pins 1, 11, 20, 30, 38 (C8–C12), USB_OTP_VDD 53 and QSPI_IOVDD 54 (C13), ADC_AVDD 44 and IOVDD 45 (C14).



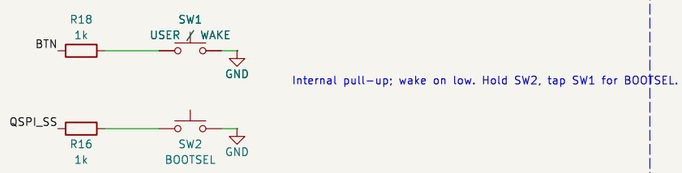
I2C0 (5x IS31FL3745, 1 MHz): 12 mA drive, LGC=1 (drivers-bga.md). LED_SDB blanks all 5 (~5 uA sleep).

ACCELEROMETER (WAKE)



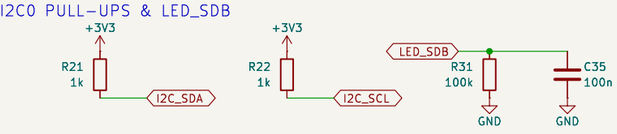
Address 0x19. CS, SAO tied high: ~20 k internal pull-ups would draw 160 uA if strapped low.

BUTTONS

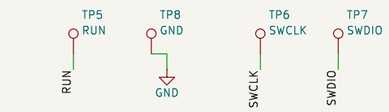


Internal pull-up; wake on low. Hold SW2, tap SW1 for BOOTSEL.

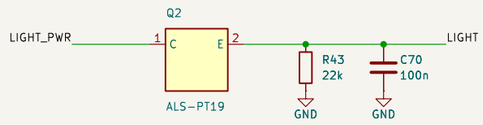
I2C0 PULL-UPS & LED_SDB



TEST PADS

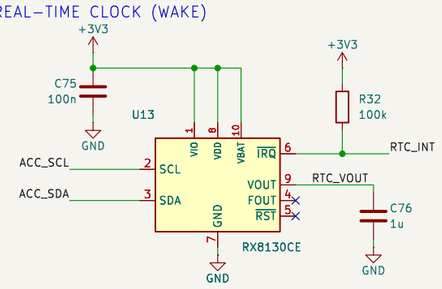


AMBIENT LIGHT SENSOR (TOP RIM)



GPIO1 powers collector (150 uA max, 0 uA in sleep). 22 k + 100 nF (tau = 2.2 ms): ~0.33 V at 100 lx, 2.9 V sat (light-sensor.md).

REAL-TIME CLOCK (WAKE)



Address 0x32 on I2C1 (400 kHz). VBAT tied to +3V3 (ETM50E-10 EX7). Set FSEL=11b (FOUT off, 1.5 uA); 100k pull-up keeps PUE=PDE=0 in sleep.

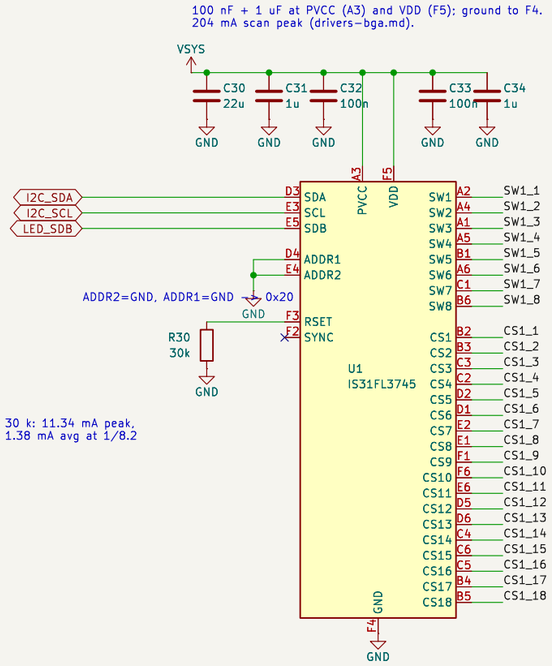
RP2350A, flash, crystal, accel, RX8130CE RTC, ALS, buttons, I2C0
Variant f-usbc-rtc (added U13 RX8130CE RTC on I2C1 + GPIO19 wake)
CERN-OHL-P-2.0

Sheet: /MCU/
File: mcu.kicad_sch

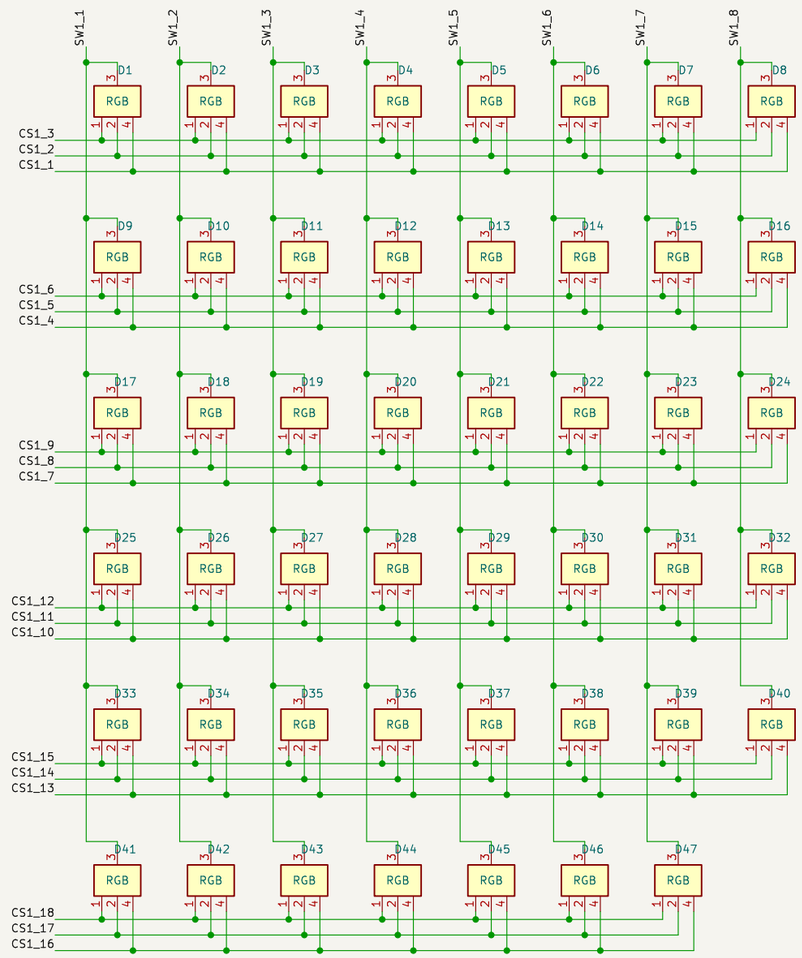
Title: round-pendant: mcu

Size: A3	Date: 2026-11-09	**Rev: FUR**
KiCad E.D.A. 9.0.8		Id: 3/8

DRIVER 1 (U1, 0x20)



47 RGB LEDS (6 x 8 − 1): R = CS(3k), G = CS(3k−1), B = CS(3k−2)



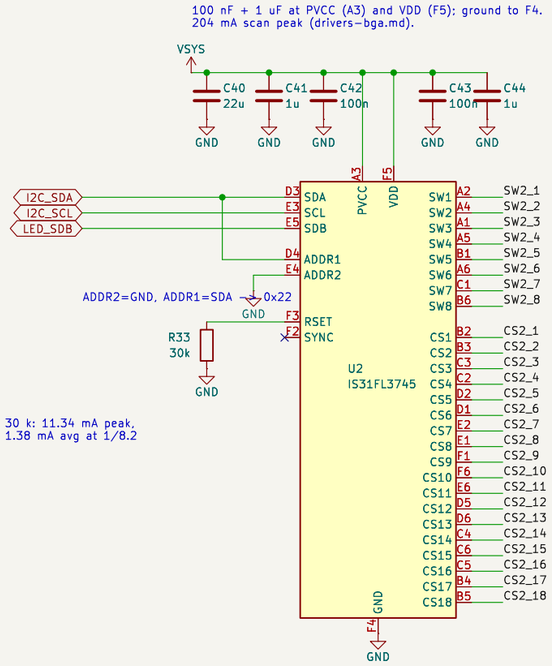
Drawn by scripts/bga/schematic_d.py
U1 IS31FL3745-CLS4-TR (0x20), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

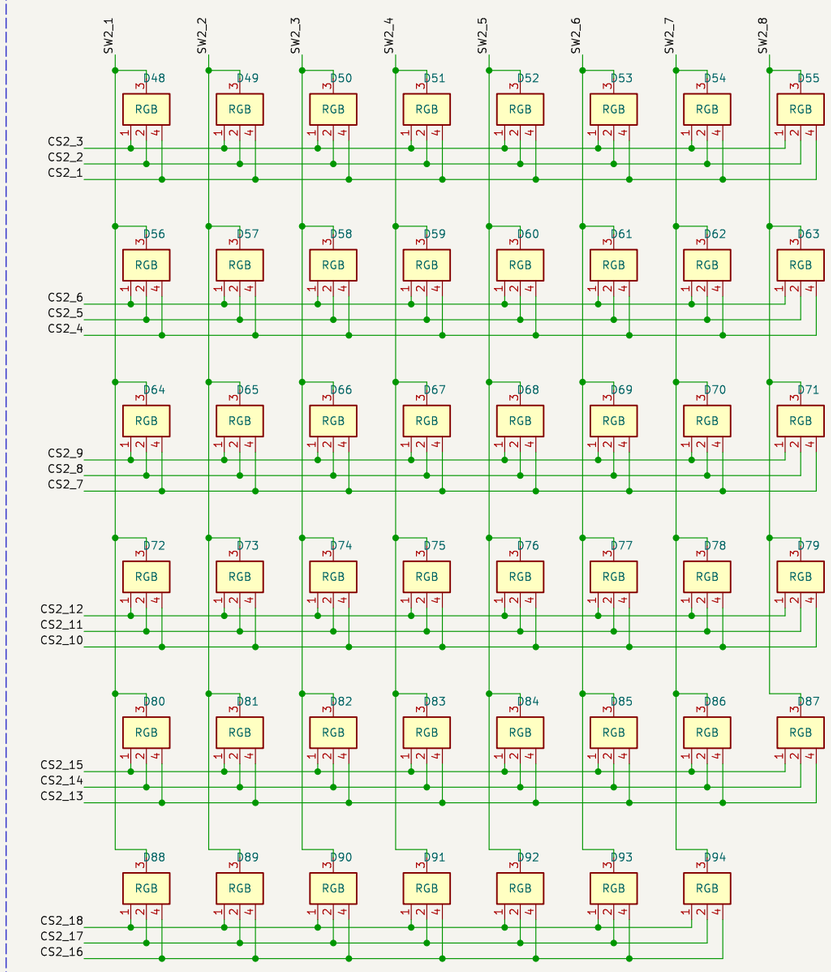
Title: round-pendant: driver 1 and matrix

Size: A3	Date: 2026-11-05	**Rev: E**
KiCad E.D.A. 9.0.8		Id: 4/8

DRIVER 2 (U2, 0x22)



47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)



Drawn by scripts/bga/schematic_d.py
U2 IS31FL3745-CLS4-TR (0x22), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

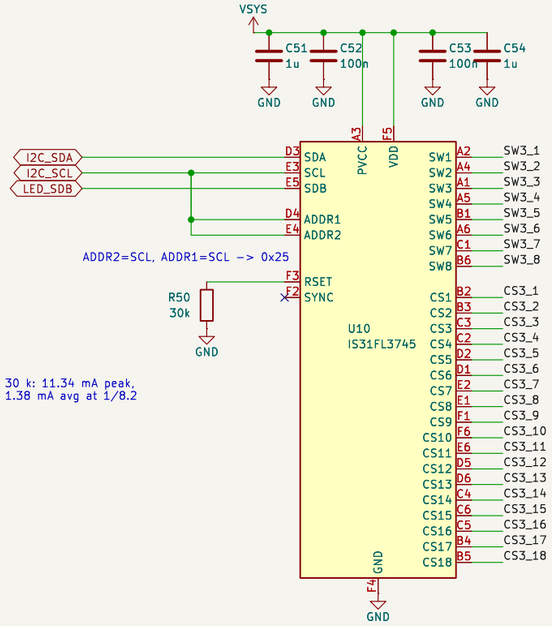
Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

Title: round-pendant: driver 2 and matrix

Size: A3	Date: 2026-11-05	**Rev: E**
KiCad E.D.A. 9.0.8		Id: 5/8

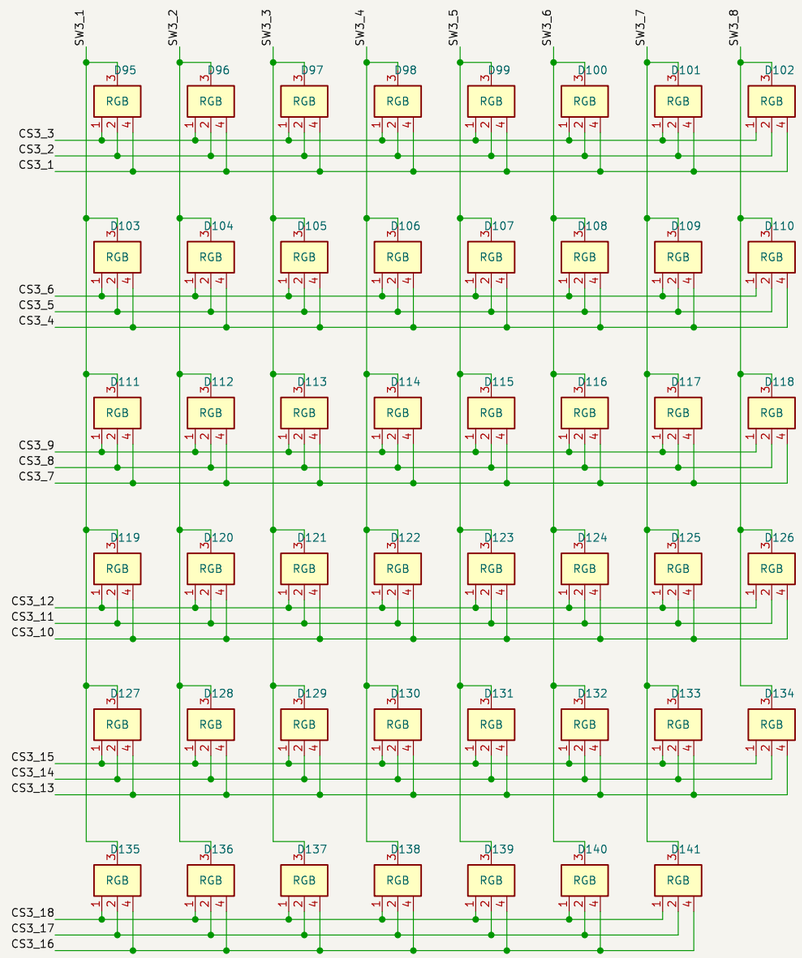
DRIVER 3 (U10, 0x25)

100 nF + 1 uF at PVCC (A3) and VDD (F5); ground to F4.
204 mA scan peak (drivers-bga.md).



30 k: 11.34 mA peak,
1.38 mA avg at 1/8.2

47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)



Drawn by scripts/bga/schematic_d.py
U10 IS31FL3745-CLS4-TR (0x25), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

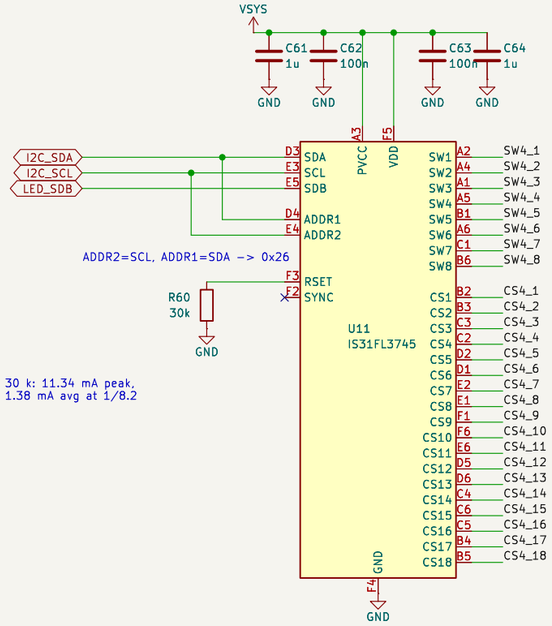
Sheet: /Driver 3 and matrix/
File: matrix3.kicad_sch

Title: round-pendant: driver 3 and matrix

Size: A3	Date: 2026-11-05	**Rev: E**
KiCad E.D.A. 9.0.8		Id: 6/8

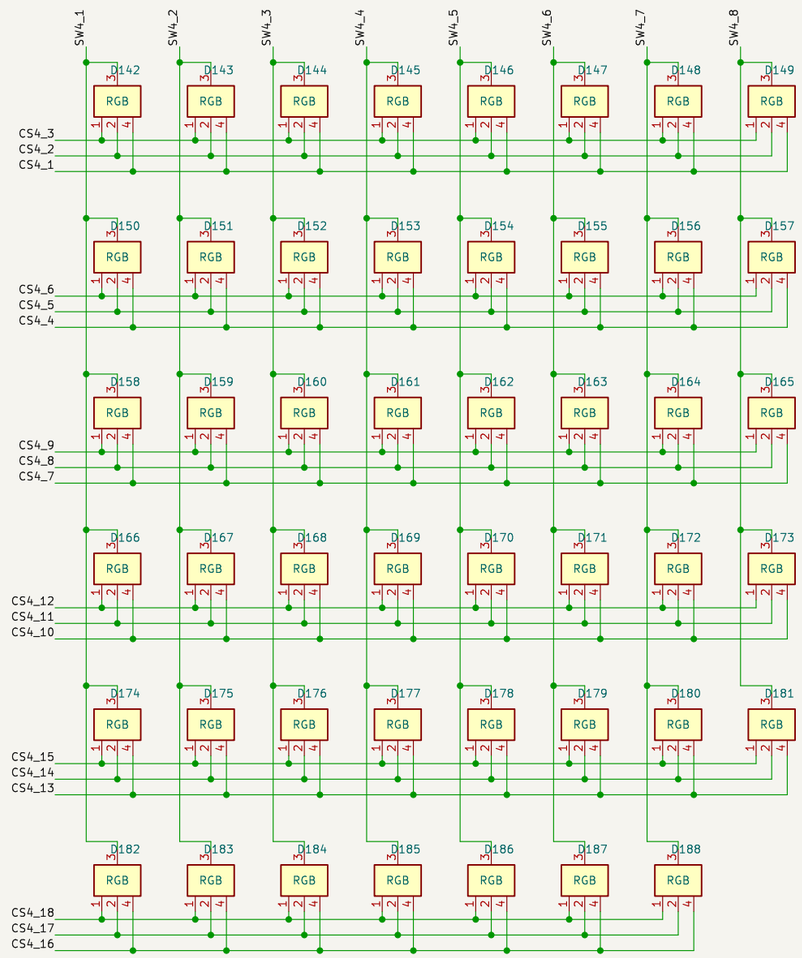
DRIVER 4 (U11, 0x26)

100 nF + 1 uF at PVCC (A3) and VDD (F5); ground to F4.
204 mA scan peak (drivers-bga.md).



30 k: 11.34 mA peak,
1.38 mA avg at 1/8.2

47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)



Drawn by scripts/bga/schematic_d.py
U11 IS31FL3745-CLS4-TR (0x26), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

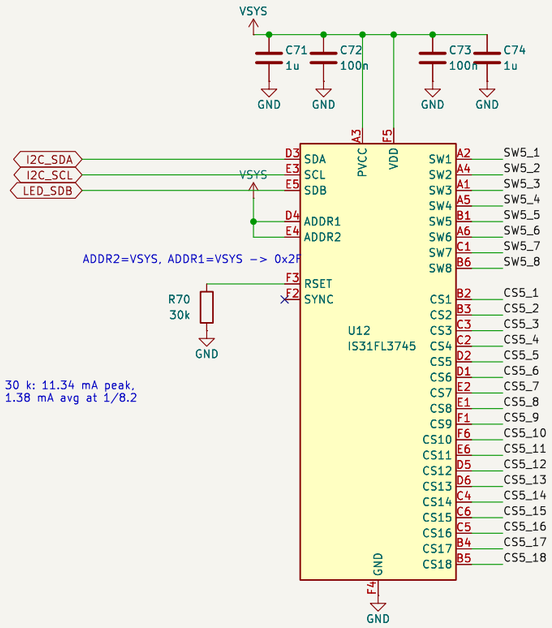
Sheet: /Driver 4 and matrix/
File: matrix4.kicad_sch

Title: round-pendant: driver 4 and matrix

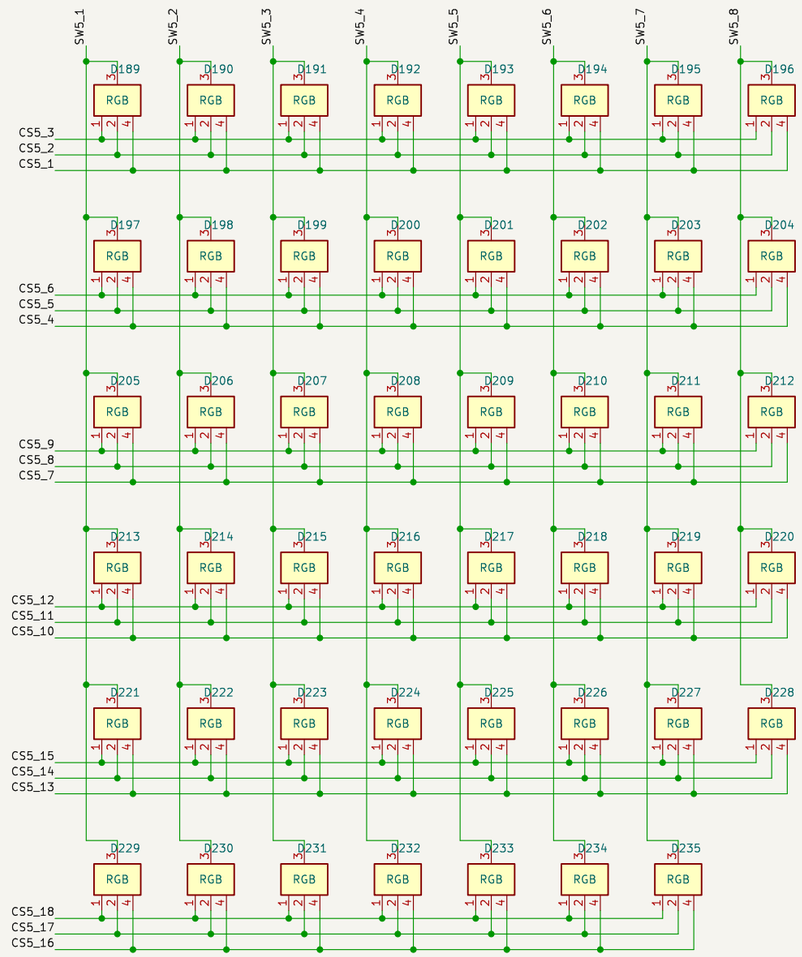
Size: A3	Date: 2026-11-05	**Rev: E**
KiCad E.D.A. 9.0.8		Id: 7/8

DRIVER 5 (U12, 0x2F)

100 nF + 1 uF at PVCC (A3) and VDD (F5); ground to F4.
204 mA scan peak (drivers-bga.md).



47 RGB LEDS (6 x 8 - 1): R = CS(3k), G = CS(3k-1), B = CS(3k-2)



Drawn by scripts/bga/schematic_d.py
U12 IS31FL3745-CLS4-TR (0x2F), 47 RGB LEDs. LED references are cell order only
CERN-OHL-P-2.0

Sheet: /Driver 5 and matrix/
File: matrix5.kicad_sch

Title: round-pendant: driver 5 and matrix

Size: A3	Date: 2026-11-05	**Rev: E**
KiCad E.D.A. 9.0.8		Id: 8/8